

THE MARGIN IN THE SHIFTED SCREW-FUNCTION CRITERION FOR ZERO-FREE HALF-PLANES

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ABSTRACT. Suzuki attached to the Riemann zeta-function a continuous real function Ψ — the screw function of ζ — and proved that the Riemann Hypothesis is equivalent to the pointwise inequality $\Psi(t) \geq 0$. He also introduced a shifted family Ψ_ω and proved that $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > \frac{1}{2} + \omega$ if and only if $\Psi_\omega(t) \geq 0$ for all sufficiently large t . The shifted family appears to have attracted no further attention.

We make the shifted criterion quantitative. Writing $\vartheta := \Theta - \frac{1}{2}$ where $\Theta = \sup\{\operatorname{Re}(\rho)\}$, we show that for $0 < \omega < \frac{1}{2}$

$$\Psi_\omega(t) = \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right)t + \left(\frac{\xi'}{\xi}\right)'\left(\frac{1}{2} + \omega\right) + O\left(e^{-(\omega - \vartheta)t}\right),$$

where the two leading coefficients are unconditional closed forms and the implied constant is explicit — under the Riemann Hypothesis it is $\omega^{-1}(\xi'/\xi)(\frac{1}{2} + \omega)$. Four consequences follow. First, the criterion carries a *margin* which is linear in t with slope exactly $(\xi'/\xi)(\frac{1}{2} + \omega)$; this slope vanishes as $\omega \rightarrow 0^+$ precisely because $(\xi'/\xi)(\frac{1}{2}) = 0$, which identifies the mechanism by which the Riemann Hypothesis is the marginal member of the family. Second, whenever the zero-free half-plane holds at all — the only case in which the criterion can hold — it holds with room to spare: $\Psi_\omega(t) > 0$ for every $t > t_0(\omega)$ with $t_0(\omega) \leq 0.0129\omega < 0.0065$, beyond which Ψ_ω admits an explicit linear lower bound. Third, a zero with $\operatorname{Re}(\rho) = \frac{1}{2} + \delta$ and $\delta > \omega$ drives Ψ_ω with an envelope growing at the exact rate $\delta - \omega$ and explicit amplitude, so that failure of the criterion is exponentially visible. Fourth, the family undergoes a sharp phase transition at $\omega_c := \Theta - \frac{1}{2}$: below ω_c the function $e^{-\varepsilon t}\Psi_\omega(t)$ oscillates between $\pm\infty$ for every $\varepsilon < \omega_c - \omega$, while at and above ω_c one has $\Psi_\omega(t) = (\xi'/\xi)(\frac{1}{2} + \omega)t + O(1)$; the Riemann Hypothesis is precisely the statement that this transition is continuous. Unconditionally, the verification of the Riemann Hypothesis to height $3 \cdot 10^{12}$, combined with the explicit zero-free regions, forces $\Psi_\omega(t) > 0$ on an explicit initial window — for $\omega = \frac{1}{4}$ up to $t = 110$, that is, $x = e^t \approx 10^{48}$ — whose endpoint obeys $t_+(\omega) = (4/c + o(1))(\frac{1}{2} - \omega)^{-2}$ with $c = 5.558691$ as $\omega \rightarrow \frac{1}{2}^-$; in particular no computation of feasible size can exhibit a violation. All results transfer to real primitive Dirichlet characters, where the margin grows with the conductor and the criterion is provably blind to real (Siegel) zeros. We verify each statement numerically, including checks of the asymptotics against the arithmetic sides of the formulas that use no information about the zeros, and close with an assessment of what the reformulation does and does not supply.

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1. INTRODUCTION

This introduction is longer and more discursive than is customary. The results below are elementary consequences of a formula of Suzuki, and their interest lies less in their proofs than in what they say about the shape of a certain family of criteria for the Riemann Hypothesis. That is hard to convey without first laying out the context, so we do.

1.1. Zeros, primes, and the explicit formula. The Riemann zeta-function $\zeta(s)$ and the prime numbers are linked by a duality made precise by Riemann and von Mangoldt. In its modern form, due to Guinand [9] and Weil [21], the link is an identity: for suitable test functions φ , with $\widehat{\varphi}(z) := \int_{\mathbb{R}} \varphi(x) e^{izx} dx$,

$$\sum_{\gamma} \widehat{\varphi}(\gamma) = \widehat{\varphi}\left(\frac{i}{2}\right) + \widehat{\varphi}\left(-\frac{i}{2}\right) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\varphi(\log n) + \varphi(-\log n)) - (\log \pi) \varphi(0) + \frac{1}{2\pi} \int_{\mathbb{R}} \operatorname{Re} \left[\frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{iz}{2} \right) \right] \widehat{\varphi}(z) dz. \quad (1)$$

Here Λ is the von Mangoldt function and the sum on the left is over the zeros γ of $\xi(\frac{1}{2} - iz)$, counted with multiplicity, where

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) \quad (2)$$

is the completed zeta-function, entire of order one and satisfying $\xi(s) = \xi(1-s)$. The Riemann Hypothesis (RH) is the assertion that every zero ρ of ξ has $\operatorname{Re}(\rho) = \frac{1}{2}$; equivalently, in the variable of (1), that every γ is real.

Identity (1) is an equality between a sum over zeros and a sum over prime powers. Read from left to right it converts spectral data into arithmetic; read from right to left it does the opposite. It is the fundamental tool for extracting information about ζ from the primes, and conversely.

1.2. Weil positivity, and the awkwardness of distributions. Weil observed that (1) encodes RH as a positivity statement. Define the *Weil distribution* W by $W(\varphi) := \sum_{\gamma} \widehat{\varphi}(\gamma)$. If every γ is real then, for $\varphi = \psi * \widetilde{\psi}$ with $\widetilde{\psi}(x) = \overline{\psi(-x)}$, one has $\widehat{\varphi}(\gamma) = |\widehat{\psi}(\gamma)|^2 \geq 0$, so $W(\psi * \widetilde{\psi}) \geq 0$; Weil proved the converse. Thus

$$\text{RH} \iff W(\psi * \widetilde{\psi}) \geq 0 \text{ for every } \psi \in C_c^{\infty}(\mathbb{R}). \quad (3)$$

This is attractive because it is *local in the primes*: if ψ is supported in $[-a, a]$, only prime powers $p^m \leq e^a$ enter the right-hand side of (1). An infinite problem about all primes becomes a family of finite problems.

It is also awkward, for a reason that is easy to state: W is a distribution, not a function. Its “kernel” is $\sum_{\gamma} e^{i\gamma t}$, which converges nowhere. The analytic work of Yoshida [23] and Bombieri [3, 4] on (3) is correspondingly technical, and the objects that arise — hermitian forms on spaces of smooth compactly supported functions with side conditions — are not the objects for which classical analysis has the most tools.

There is a second structural difficulty. One would like to test (3) on a small family of ψ rather than on all of C_c^∞ , and it is a priori unclear that any small family suffices. That it can is the content of Li's criterion [13], which Bombieri and Lagarias [5] showed to be exactly the positivity of W on a distinguished countable family of test functions. This is worth emphasising, since it is the precedent for everything below: *RH is already equivalent to Weil positivity on a one-parameter family.*

1.3. Integrating twice: the screw function. Suzuki's contribution [17] is to integrate the Weil distribution twice. Set

$$\Psi(t) := 4(e^{t/2} + e^{-t/2} - 2) - \sum_{n \leq e^t} \frac{\Lambda(n)}{\sqrt{n}} (t - \log n) + \frac{t}{2} \left[\frac{\Gamma'}{\Gamma} \left(\frac{1}{4} \right) - \log \pi \right] + \frac{1}{4} \left(C - e^{-t/2} \Phi(e^{-2t}, 2, \frac{1}{4}) \right) \quad (4)$$

for $t \geq 0$, extended to an even function of $t \in \mathbb{R}$; here Φ is the Hurwitz–Lerch zeta-function and $C = \zeta(2, \frac{1}{4})$. Then $\Psi'' = W$ as distributions, and Ψ is a *continuous* real function with the two faces

$$\Psi(t) = \sum_{\gamma} \frac{1 - \cos(\gamma t)}{\gamma^2}, \quad \int_0^\infty \Psi(t) e^{izt} dt = -\frac{1}{z^2} \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} - iz \right) \quad (\text{Im } z > \frac{1}{2}). \quad (5)$$

Two observations make Ψ concrete. First, by the prime number theorem the prime sum in (4) equals $4e^{t/2} + O(e^{t/2} e^{-c\sqrt{t}})$ [17, Prop. 2.1], so the leading exponential cancels against the first term: *Ψ is the error term in the prime number theorem, twice integrated and normalised.* Second, if RH holds then every term of the series in (5) is non-negative and Ψ is confined to a narrow band, $0 \leq \Psi(t) < 2 \sum_{\gamma} \gamma^{-2} < 0.0925$.

The resulting criterion [17, Thm. 1.7] is as concrete as any known form of RH:

$$\text{RH} \iff \Psi(t) \geq 0 \text{ for every } t. \quad (6)$$

Necessity is immediate from (5). Sufficiency does not use Weil's argument at all: it follows from the classical theorem that the Laplace transform of a non-negative function has a singularity at the real point of its abscissa of convergence [22, Ch. II, Thm. 5b], applied to the second identity in (5), together with the elementary fact that ζ has no zeros on $[\frac{1}{2}, \infty)$.

Since $\Psi(t) = W(\Delta_t)$ where $\Delta_t = R_t * \widetilde{R}_t$ is the triangular function supported on $[-t, t]$ and $R_t = 2^{-1/2} \mathbf{1}_{[-t/2, t/2]}$ a normalised rectangular function [17, §3.4], (6) says that Weil positivity tested on the family of *rectangular* functions alone already characterises RH. As noted above, the existence of such a reduction is not new — Li's criterion is one — but the family here is a continuum of compactly supported functions rather than a countable family, and the resulting object Ψ is a continuous function of one real variable with an explicit arithmetic formula. That is what makes it possible to attach the classical theory of Kreĭn [11], in which $-\Psi$ is a *screw function*: a continuous hermitian g with $g(t-u) - g(t) - g(-u) + g(0)$ a non-negative kernel. Equivalently, $-g$ is conditionally negative definite, so the natural home of the positivity is the increments of Ψ , not Ψ itself.

1.4. Why RH is the hard member of a family. Criterion (6) is exact, and exactly saturated. Suzuki observes [17, §7.1] that if RH holds, the zeros are simple, and their positive ordinates are linearly independent over $\mathbb{Q}$, then Kronecker's theorem gives

$$\limsup_{t \rightarrow \infty} \Psi(t) = 2 \sum_{\gamma} \frac{1}{\gamma^2}, \quad \liminf_{t \rightarrow \infty} \Psi(t) = 0. \quad (7)$$

The function comes arbitrarily close to violating the inequality, infinitely often. As Suzuki puts it, this “suggests the difficulty of proving the non-negativity of $\Psi(t)$ by approximation.”

There is, however, a graded version. Following Lagarias [12], whose Theorem 1.1 is already stated with a general abscissa θ , Suzuki introduces [17, §11] for $\omega \in \mathbb{R}$ the shifted family

$$\Psi_\omega(t) := e^{-\omega t} \Psi(t) + 2\omega \int_0^t e^{-\omega u} \Psi(u) du + \omega^2 \int_0^t (t-u) e^{-\omega u} \Psi(u) du \quad (t > 0), \quad (8)$$

again extended evenly, so that $\Psi_0 = \Psi$ and

$$\int_0^\infty \Psi_\omega(t) e^{izt} dt = -\frac{1}{z^2} \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega - iz \right), \quad \text{Im } z > \frac{1}{2} - \omega, \quad (9)$$

and proves:

Theorem 1.1 (Suzuki [17, Thm. 11.1]). *Let $\omega \in \mathbb{R}$. Then $\xi(s) \neq 0$ for $\text{Re}(s) > \frac{1}{2} + \omega$ if and only if there exists $t_0 > 0$ such that $\Psi_\omega(t) \geq 0$ for all $t \geq t_0$.*

Two remarks of Suzuki’s about Theorem 1.1 matter for what follows. First, in the necessity direction the qualifier “sufficiently large” can be dispensed with: Suzuki observes [17, §11] that if $\xi(s) \neq 0$ for $\text{Re}(s) > \frac{1}{2} + \omega$ then $i(\xi'/\xi)(\frac{1}{2} + \omega - iz)$ is a Herglotz function, so $-\Psi_\omega$ is a screw function by the Kreĭn–Langer correspondence, and $\Psi_\omega(t) \geq 0$ for *every* t . The weakening to “eventually” strengthens only the sufficiency direction, where it weakens the hypothesis. Second, the family is monotone: $\Psi_{\omega+\eta}$ is obtained from Ψ_ω by the same transformation (8) with parameter η , so non-negativity propagates upward in ω .

Writing $\Theta := \sup\{\text{Re}(\rho) : \xi(\rho) = 0\}$ and $\omega_c := \inf\{\omega : \Psi_\omega \geq 0 \text{ eventually}\}$, Theorem 1.1 says $\omega_c = \Theta - \frac{1}{2}$, and RH is the case $\omega_c = 0$. Two things should be said about this range. For $\omega \geq \frac{1}{2}$ the criterion is vacuous, since $\zeta(s) \neq 0$ for $\text{Re}(s) \geq 1$ is equivalent to the prime number theorem. For every $\omega < \frac{1}{2}$ it is open: all known zero-free regions have the shape $\sigma \geq 1 - f(|t|)$ with $f(|t|) \rightarrow 0$ — the Korobov–Vinogradov form [8], or in explicit form $\sigma \geq 1 - (55.241(\log |t|)^{2/3}(\log \log |t|)^{1/3})^{-1}$ [14] — and none of them contains a half-plane $\sigma > c$ with $c < 1$. Whether $\Theta < 1$ is not known.

So Theorem 1.1 interpolates, between a vacuous endpoint and RH, through a range of statements every one of which is open. That makes it worth asking how the criterion behaves as a function of ω , and in particular whether the saturation (7) that obstructs the case $\omega = 0$ persists for $\omega > 0$. That is the question this note answers.

1.5. Results. Throughout, $\vartheta := \Theta - \frac{1}{2} \in [0, \frac{1}{2}]$, and we write

Hypothesis $\mathbf{H}(\vartheta)$. $\xi(s) \neq 0$ for $\text{Re}(s) > \frac{1}{2} + \vartheta$.

This holds by the definition of ϑ ; RH is the case $\vartheta = 0$. Define

$$A(\omega) := \sum_{\gamma} \frac{1}{\gamma^2 + \omega^2}, \quad B(\omega) := \sum_{\gamma} \frac{\gamma^2 - \omega^2}{(\gamma^2 + \omega^2)^2}, \quad A^*(\omega) := \frac{1}{2} \sum_{\gamma} \left[\frac{1}{|\gamma - i\omega|^2} + \frac{1}{|\gamma + i\omega|^2} \right]. \quad (10)$$

Our first observation is that the first two sums are classical objects in disguise, with no hypothesis at all.

Proposition 1.2. *For $0 < \omega < \frac{1}{2}$ the series (10) converge absolutely, and, unconditionally,*

$$A(\omega) = \frac{1}{\omega} \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right), \quad B(\omega) = \left(\frac{\xi'}{\xi} \right)' \left(\frac{1}{2} + \omega \right).$$

Moreover $A(\omega)$ is strictly decreasing on $(0, \frac{1}{2})$ with $\lim_{\omega \rightarrow \frac{1}{2}^-} A(\omega) = \sum_{\rho} \frac{1}{\rho(1-\rho)} = 2 + \gamma_0 - \log 4\pi = 0.0461914\dots$, where γ_0 is Euler's constant. If RH holds, $A^*(\omega) = A(\omega)$.

The main result is then a two-term asymptotic with explicit error. Only the error term is conditional.

Theorem 1.3. *Let $0 < \omega < \frac{1}{2}$ and assume $\vartheta \leq \omega$. Then for all $t \geq 0$,*

$$\Psi_{\omega}(t) = \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right)t + \left(\frac{\xi'}{\xi}\right)'\left(\frac{1}{2} + \omega\right) + R_{\omega}(t), \quad |R_{\omega}(t)| \leq A^*(\omega)e^{-(\omega-\vartheta)t}.$$

In particular, under RH, $|R_{\omega}(t)| \leq \omega^{-1}(\xi'/\xi)(\frac{1}{2} + \omega)e^{-\omega t}$.

The linear term is what we call the *margin* of the criterion, and its slope is $(\xi'/\xi)(\frac{1}{2} + \omega)$. Since $\xi(s) = \xi(1-s)$ forces $(\xi'/\xi)(\frac{1}{2}) = 0$, the margin vanishes as $\omega \rightarrow 0^+$, at the rate $(\xi'/\xi)'(\frac{1}{2})\omega = 0.0462099862\dots \times \omega$. This identifies the mechanism behind (7): *the marginality of the Riemann Hypothesis inside this family is exactly the vanishing of ξ'/ξ at the central point*.

The margin also shows that whenever the criterion of Theorem 1.1 holds at all, it holds with an explicit strict margin from an explicit, and very small, threshold onward.

Theorem 1.4. *Let $0 < \omega < \frac{1}{2}$ and assume $\vartheta \leq \omega$, i.e. $\xi(s) \neq 0$ for $\operatorname{Re}(s) > \frac{1}{2} + \omega$ — precisely the situation in which the criterion holds. Then for every $t \geq 0$,*

$$\Psi_{\omega}(t) \geq \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right)t + \left(\frac{\xi'}{\xi}\right)'\left(\frac{1}{2} + \omega\right) - A^*(\omega),$$

and $\Psi_{\omega}(t) > 0$ for every $t > t_0(\omega) := (A^(\omega) - B(\omega))/(\omega A(\omega))$. Moreover*

$$A^*(\omega) - B(\omega) \leq 8\omega^2 \sum_{\rho} \frac{1}{(\operatorname{Im} \rho)^4}, \quad \text{whence} \quad t_0(\omega) \leq \frac{8 \sum_{\rho} (\operatorname{Im} \rho)^{-4}}{2 + \gamma_0 - \log 4\pi} \omega = 0.01288\dots \times \omega < 0.0065.$$

The two regimes — linear growth at and above $\omega_c := \Theta - \frac{1}{2}$, failure below — meet in a sharp transition, and the behaviour below ω_c is not merely a failure of non-negativity but two-sided exponential oscillation.

Theorem 1.5. *Let $0 < \omega < \frac{1}{2}$.*

- (i) *If $\vartheta \leq \omega$ then $\Psi_{\omega}(t) = (\xi'/\xi)(\frac{1}{2} + \omega)t + O(1)$ with $|O(1)| \leq |B(\omega)| + A^*(\omega)$; in particular $\Psi_{\omega}(t)/t \rightarrow (\xi'/\xi)(\frac{1}{2} + \omega)$.*
- (ii) *If $\omega < \vartheta$ then for every $0 \leq \varepsilon < \vartheta - \omega$,*

$$\limsup_{t \rightarrow \infty} e^{-\varepsilon t} \Psi_{\omega}(t) = +\infty, \quad \liminf_{t \rightarrow \infty} e^{-\varepsilon t} \Psi_{\omega}(t) = -\infty.$$

Corollary 1.6. $\Theta \leq \frac{1}{2} + \omega$ *if and only if $\Psi_{\omega}(t) = (\xi'/\xi)(\frac{1}{2} + \omega)t + O(1)$. Consequently $m(\omega) := \liminf_{t \rightarrow \infty} \Psi_{\omega}(t)/t$ equals $-\infty$ on $(0, \omega_c)$ and $(\xi'/\xi)(\frac{1}{2} + \omega)$ on $[\omega_c, \frac{1}{2})$: the normalised slope is an order parameter, with a jump of size at least $(2 + \gamma_0 - \log 4\pi)\omega_c$ at ω_c . The Riemann Hypothesis holds if and only if the transition is continuous, i.e. $\omega_c = 0$ and $m(\omega) \rightarrow 0$ as $\omega \rightarrow 0^+$.*

Corollary 1.6 is a graded form of Suzuki's Theorem 1.6 ($\text{RH} \Leftrightarrow \Psi = O(1)$), with the sharp linear term; it also yields yet another description of the critical shift: $\omega_c = \inf\{\omega : \Psi_{\omega}(t) - (\xi'/\xi)(\frac{1}{2} + \omega)t \text{ is bounded}\}$.

Next, failure of the criterion is exponentially visible, at a rate that reads off the depth of the violation.

Proposition 1.7. *Let $\rho_0 = \frac{1}{2} + \delta + i\tau_0$ be a zero of ξ with $\delta > \omega \geq 0$ and $\tau_0 > 0$, and let $\Psi_\omega^{(\rho_0)}$ denote the contribution to the series for Ψ_ω of the quartet $\{\rho_0, \bar{\rho}_0, 1 - \rho_0, 1 - \bar{\rho}_0\}$. Then*

$$\Psi_\omega^{(\rho_0)}(t) = -\frac{2e^{(\delta-\omega)t}}{\tau_0^2 + (\delta-\omega)^2} \cos(\tau_0 t + \alpha) + O(t), \quad \alpha = 2 \arg(\tau_0 + i(\delta - \omega));$$

in particular $\limsup_{t \rightarrow \infty} e^{-(\delta-\omega)t} |\Psi_\omega^{(\rho_0)}(t)| = 2(\tau_0^2 + (\delta - \omega)^2)^{-1} > 0$.

Combining the sensitivity estimate with the numerical verification of RH to height $3 \cdot 10^{12}$ [16] gives an unconditional initial window of positivity, which we record because it delimits sharply what computation can and cannot say about the criterion.

Proposition 1.8. *Let $0 < \omega < \frac{1}{2}$ and $T_v = 3 \cdot 10^{12}$. Unconditionally, for every $t \geq 0$,*

$$\Psi_\omega(t) \geq \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right)t + \left(\frac{\xi'}{\xi}\right)'\left(\frac{1}{2} + \omega\right) - A(\omega)e^{-\omega t} - e^{(\frac{1}{2}-\omega)t}E(t),$$

where, with $L_v := \log T_v$ and $\nu(L) := \max\{(5.558691 L)^{-1}, (55.241 L^{2/3}(\log L)^{1/3})^{-1}\}$ the explicit zero-free function of [14],

$$E(t) := \frac{1}{\pi} \int_{L_v}^{\infty} (L - \log 2\pi) e^{-L - \nu(L)t} dL, \quad E(t) \leq E(0) = S := \sum_{|\operatorname{Im} \rho| > T_v} \frac{1}{(\operatorname{Im} \rho)^2} < 3.2 \cdot 10^{-12}.$$

Consequently $\Psi_\omega(t) > 0$ on an explicit window $t_-(\omega) < t < t_+(\omega)$ with $t_-(\omega) < 0.0017\omega$ and, for example, $t_+(0.1) = 64.7$, $t_+(0.25) = 109.9$, $t_+(0.4) = 300.9$, $t_+(0.49) = 7262$; as $\omega \rightarrow \frac{1}{2}^-$,

$$t_+(\omega) = \left(\frac{4}{c} + o(1)\right) \left(\frac{1}{2} - \omega\right)^{-2}, \quad c = 5.558691.$$

All of this transfers to real primitive Dirichlet characters, with two instructive changes (Section 9). The closed forms, the margin, the threshold and the transition hold verbatim for the shifted screw functions $\Psi_{\omega, \chi}$, with slope $(\xi'/\xi)(\frac{1}{2} + \omega, \chi)$ vanishing at $\omega = 0$ for the same functional-equation reason — and the margin *grows* with the conductor ($B_\chi(0) = 0.113, 0.156, 0.157$ for $q = 3, 4, 5$, against 0.0462 for ζ): within this family, ζ is the most marginal member. The arithmetic side is Proposition 9.2. The one genuinely new phenomenon is negative: the criterion is blind to real zeros — a Siegel zero pushes $\Psi_{\omega, \chi}$ *up*, exponentially (Proposition 9.3) — so what the shifted criterion characterises for χ is the location of the *non-real* zeros only. This explains quantitatively the real-zero hypothesis in Suzuki's extension of the screw-function theory to the extended Selberg class [18, Thm. 1.1].

Sections 2–8 prove these statements; Section 9 carries them to Dirichlet L -functions and discusses the extended Selberg class; Section 10 verifies them numerically, including a test of Theorem 1.3 against the arithmetic side of (4) which uses no information about the zeros; Section 11 assesses what the reformulation supplies.

Attribution. The function Ψ , the shifted family Ψ_ω , Theorem 1.1, and the observation that a zero-free half-plane forces $\Psi_\omega \geq 0$ pointwise are Suzuki's [17]; the screw-function theory over the extended Selberg class, including the role of real zeros, is developed by him in [18]. The graded criterion in the form “ $\operatorname{Re}(\rho) \leq \theta$ for all ρ if and only if a certain function is Herglotz on $\operatorname{Re}(s) > \theta$ ” is Lagarias's [12, Thm. 1.1]. The reduction of RH to Weil positivity on a one-parameter family of test functions goes back to Li [13] and Bombieri–Lagarias [5]. What is offered here is Proposition 1.2, Theorems 1.3, 1.4 and 1.5, Corollary 1.6, Propositions 1.7 and 1.8, the Dirichlet and Selberg-class material of Section 9 (Theorem 9.1, Propositions 9.2 and 9.3), the numerical verification, and the discussion in Section 11.

2. SETUP

We use Suzuki's normalisation throughout. Let $\Xi(z) := \xi(\frac{1}{2} - iz)$, an entire function of order one; by $\xi(s) = \xi(1 - s)$ it is even, and its zeros γ are related to the zeros ρ of ξ by

$$\gamma = i\left(\rho - \frac{1}{2}\right), \quad \text{i.e.} \quad \rho = \beta + i\tau \implies \gamma = -\tau + i\left(\beta - \frac{1}{2}\right). \quad (11)$$

Thus $\text{Im } \gamma = \text{Re } \rho - \frac{1}{2}$, RH is the statement that all γ are real, and Hypothesis $\text{H}(\vartheta)$ says $|\text{Im } \gamma| \leq \vartheta$ for every γ . Zeros come in the symmetric configuration $\gamma \mapsto -\gamma$, $\gamma \mapsto \bar{\gamma}$. Two unconditional facts about the zeros are used repeatedly: $|\text{Im } \gamma| \leq \frac{1}{2}$ for every γ [19, Thm. 2.12], and $|\text{Re } \gamma| \geq 14$, since the nontrivial zero of least height has ordinate 14.1347... [19, §15].

We shall use the standard Hadamard consequences (see e.g. [7, §12])

$$\frac{\Xi'}{\Xi}(z) = \sum_{\gamma} \frac{1}{z - \gamma}, \quad \frac{\Xi'}{\Xi}(z) = -i \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} - iz\right), \quad \left(\frac{\xi'}{\xi}\right)(1 - s) = -\left(\frac{\xi'}{\xi}\right)(s), \quad (12)$$

the first with the usual convention that γ and $-\gamma$ are paired.

The starting point is Suzuki's series for the shifted family [17, §11]: for $t > 0$,

$$\Psi_{\omega}(t) = \sum_{\gamma} \frac{1}{(\gamma^2 + \omega^2)^2} \left[\omega t(\gamma^2 + \omega^2) + (\gamma^2 - \omega^2) - e^{-\omega t}(\gamma^2 - \omega^2) \cos(\gamma t) - e^{-\omega t} 2\gamma\omega \sin(\gamma t) \right]. \quad (13)$$

Setting $\omega = 0$ recovers $\Psi(t) = \sum_{\gamma} \gamma^{-2}(1 - \cos \gamma t)$, as it must. Suzuki also gives the arithmetic counterpart, which we record because it is what we compute with in Section 10: with $\psi = \Gamma'/\Gamma$ and $\psi^{(1)} = \psi'$,

$$\begin{aligned} \Psi_{\omega}(t) = 4 \left[\frac{e^{(\frac{1}{2}-\omega)t}}{(1-2\omega)^2} + \frac{e^{-(\frac{1}{2}+\omega)t}}{(1+2\omega)^2} - \frac{4-2(1-\omega t)(1-4\omega^2)}{(1-4\omega^2)^2} \right] + \frac{t}{2} \left[\psi\left(\frac{1}{4} + \frac{\omega}{2}\right) - \log \pi \right] \\ + \frac{1}{4} \left[\psi^{(1)}\left(\frac{1}{4} + \frac{\omega}{2}\right) - e^{-(\frac{1}{2}+\omega)t} \Phi(e^{-2t}, 2, \frac{1}{4} + \frac{\omega}{2}) \right] - \sum_{n \leq e^t} \frac{\Lambda(n)}{n^{\frac{1}{2}+\omega}} (t - \log n). \end{aligned} \quad (14)$$

Lemma 2.1. *Let $0 < \omega < \frac{1}{2}$. Unconditionally, $\text{Re}(\gamma^2 + \omega^2) \geq (\text{Re } \gamma)^2 - \frac{1}{4} + \omega^2 > 0$ for every γ , the series (10) converge absolutely, and $\sum_{\gamma} |\gamma| |\gamma^2 + \omega^2|^{-2} < \infty$.*

Proof. Write $\gamma = a + ib$; then $|b| \leq \frac{1}{2}$ and $|a| \geq 14$. We have $\gamma^2 + \omega^2 = a^2 - b^2 + \omega^2 + 2iab$, whose real part is at least $a^2 - \frac{1}{4} > 0$. Hence $|\gamma^2 + \omega^2| \geq a^2 - \frac{1}{4}$, and likewise $|\gamma \pm i\omega|^2 = a^2 + (b \pm \omega)^2 \geq a^2$. By the Riemann–von Mangoldt formula the number of γ with $|a| \leq T$ is $O(T \log T)$, so $\sum_{\gamma} (a^2 - \frac{1}{4})^{-1} < \infty$ and $\sum_{\gamma} |\gamma| (a^2 - \frac{1}{4})^{-2} < \infty$ by partial summation. The stated bounds follow. $\square$

Remark 2.2. Suzuki also records the composition law: $\Psi_{\omega+\eta}$ arises from Ψ_{ω} by the transformation (8) with parameter $\eta > 0$ [17, §11]. Since that transformation has non-negative kernel, non-negativity of Ψ_{ω} (pointwise, or on $[0, t_1]$) implies the same for $\Psi_{\omega'}$, $\omega' > \omega$: the family is monotone in ω .

3. TWO SPECTRAL SUMS IN CLOSED FORM

Proof of Proposition 1.2. Convergence is Lemma 2.1, and requires no hypothesis. For the closed forms, evaluate the first identity of (12) at $z = \pm i\omega$ (legitimate for $0 < \omega < \frac{1}{2}$ since no γ lies on the imaginary axis: $|\text{Re } \gamma| \geq 14$) and use the second:

$$\sum_{\gamma} \frac{1}{i\omega - \gamma} = -i \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right), \quad \sum_{\gamma} \frac{1}{-i\omega - \gamma} = -i \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} - \omega\right) = i \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right),$$

the last step by the functional equation in (12). Therefore

$$\sum_{\gamma} \frac{1}{\gamma - i\omega} = i \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right), \quad \sum_{\gamma} \frac{1}{\gamma + i\omega} = -i \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right).$$

Since $\gamma^2 + \omega^2 = (\gamma - i\omega)(\gamma + i\omega)$, partial fractions give

$$A(\omega) = \sum_{\gamma} \frac{1}{(\gamma - i\omega)(\gamma + i\omega)} = \frac{1}{2i\omega} \sum_{\gamma} \left[\frac{1}{\gamma - i\omega} - \frac{1}{\gamma + i\omega} \right] = \frac{1}{2i\omega} \cdot 2i \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right) = \frac{1}{\omega} \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right),$$

the rearrangement being justified by absolute convergence of $A(\omega)$ together with the standard pairing of γ with $-\gamma$ in (12).

For B , note the elementary identity

$$\frac{\partial}{\partial \omega} \left[\frac{\omega}{\gamma^2 + \omega^2} \right] = \frac{\gamma^2 - \omega^2}{(\gamma^2 + \omega^2)^2}.$$

By Lemma 2.1 the differentiated series converges locally uniformly on $0 < \omega < \frac{1}{2}$, so term-by-term differentiation of $\omega A(\omega)$ is legitimate and

$$B(\omega) = \frac{d}{d\omega} [\omega A(\omega)] = \frac{d}{d\omega} \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right) = \left(\frac{\xi'}{\xi} \right)' \left(\frac{1}{2} + \omega \right).$$

For monotonicity, pair each γ with $\bar{\gamma}$, so that $A(\omega) = \sum_{\gamma} X_{\gamma} / (X_{\gamma}^2 + Y_{\gamma}^2)$ with $X_{\gamma} = a^2 - b^2 + \omega^2 > 0$ and $Y_{\gamma} = 2ab$. Then

$$\frac{\partial}{\partial(\omega^2)} \frac{X}{X^2 + Y^2} = \frac{Y^2 - X^2}{(X^2 + Y^2)^2} < 0,$$

since $X > |Y|$ is equivalent to $(|a| - |b|)^2 + \omega^2 > 0$. So every paired term, hence $A(\omega)$, is strictly decreasing. At $\omega = \frac{1}{2}$, $\gamma^2 + \frac{1}{4} = \rho(1 - \rho)$ by (11), and $\sum_{\rho} 1/(\rho(1 - \rho)) = 2 \sum_{\rho} 1/\rho = 2 + \gamma_0 - \log 4\pi$ is classical [7, §12]; the limit follows by dominated convergence. Finally, if RH holds then γ is real, $|\gamma \pm i\omega|^2 = \gamma^2 + \omega^2$, and $A^* = A$. $\square$

Remark 3.1. Letting $\omega \rightarrow 0^+$ gives $B(0) = \sum_{\gamma} \gamma^{-2} = (\xi'/\xi)'(\frac{1}{2}) = 0.0462099862308\dots$ (an unconditional identity), so $2B(0) = 0.0924199725\dots$, consistent with Suzuki's conditional bound $\sup \Psi < 0.094$. Also $A(\omega) \rightarrow B(0)$ as $\omega \rightarrow 0^+$, since $(\xi'/\xi)(\frac{1}{2}) = 0$. Note the coincidence of scales: A decreases only from 0.046210 to 0.046191 across the whole interval $(0, \frac{1}{2})$.

4. THE ASYMPTOTIC

Proof of Theorem 1.3. Split (13) into the two groups of terms. The first two terms sum, by (10) and Proposition 1.2, to

$$\omega t \sum_{\gamma} \frac{1}{\gamma^2 + \omega^2} + \sum_{\gamma} \frac{\gamma^2 - \omega^2}{(\gamma^2 + \omega^2)^2} = \omega A(\omega) t + B(\omega) = \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega \right) t + \left(\frac{\xi'}{\xi} \right)' \left(\frac{1}{2} + \omega \right),$$

which is the stated main term, and is unconditional. The remainder is

$$R_{\omega}(t) = -e^{-\omega t} \sum_{\gamma} \frac{(\gamma^2 - \omega^2) \cos(\gamma t) + 2\gamma\omega \sin(\gamma t)}{(\gamma^2 + \omega^2)^2}. \quad (15)$$

The key is the exact decomposition

$$(\gamma^2 - \omega^2) \cos(\gamma t) + 2\gamma\omega \sin(\gamma t) = \frac{1}{2} e^{i\gamma t} (\gamma - i\omega)^2 + \frac{1}{2} e^{-i\gamma t} (\gamma + i\omega)^2, \quad (16)$$

which is immediate from $\cos w = \frac{1}{2}(e^{iw} + e^{-iw})$, $\sin w = \frac{1}{2i}(e^{iw} - e^{-iw})$ and $(\gamma \mp i\omega)^2 = \gamma^2 - \omega^2 \mp 2i\gamma\omega$. Dividing by $(\gamma^2 + \omega^2)^2 = (\gamma - i\omega)^2(\gamma + i\omega)^2$,

$$R_\omega(t) = -\frac{e^{-\omega t}}{2} \sum_{\gamma} \left[\frac{e^{i\gamma t}}{(\gamma + i\omega)^2} + \frac{e^{-i\gamma t}}{(\gamma - i\omega)^2} \right]. \quad (17)$$

Writing $\gamma = a + ib$ with $|b| \leq \vartheta$, we have $|e^{\pm i\gamma t}| = e^{\mp bt} \leq e^{\vartheta t}$, whence

$$|R_\omega(t)| \leq \frac{e^{-(\omega-\vartheta)t}}{2} \sum_{\gamma} \left[\frac{1}{|\gamma + i\omega|^2} + \frac{1}{|\gamma - i\omega|^2} \right] = A^*(\omega) e^{-(\omega-\vartheta)t}.$$

Under RH, $A^* = A$ by Proposition 1.2, giving the stated constant. $\square$

Remark 4.1. The one-line amplitude computation behind (16) — for real γ , $(\gamma^2 - \omega^2) \cos \theta + 2\gamma\omega \sin \theta$ has amplitude exactly $\sqrt{(\gamma^2 - \omega^2)^2 + 4\gamma^2\omega^2} = \gamma^2 + \omega^2$ — shows the constant $A(\omega)$ in the RH case is the best possible for a term-by-term estimate. Note also that the bound holds at $\vartheta = \omega$: there the remainder no longer decays, but stays bounded by $A^*(\omega)$, which is all Theorem 1.4 needs.

Corollary 4.2. *The slope of the linear term in Theorem 1.3 is $(\xi'/\xi)(\frac{1}{2} + \omega)$. Unconditionally, it is positive for $0 < \omega < \frac{1}{2}$, indeed*

$$\left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right) = \omega A(\omega) \geq (2 + \gamma_0 - \log 4\pi) \omega = 0.0461914 \dots \times \omega,$$

it vanishes at $\omega = 0$, and it satisfies $(\xi'/\xi)(\frac{1}{2} + \omega) = B(0)\omega + O(\omega^3)$ with $B(0) = 0.04620998623 \dots$

Proof. $\xi(s) = \xi(1-s)$ gives $(\xi'/\xi)(1-s) = -(\xi'/\xi)(s)$, so (ξ'/ξ) is odd about $s = \frac{1}{2}$; in particular $(\xi'/\xi)(\frac{1}{2}) = 0$ and its Taylor expansion there contains only odd powers. The linear coefficient is $(\xi'/\xi)'(\frac{1}{2}) = B(0)$ by Remark 3.1. The lower bound is the monotonicity statement of Proposition 1.2. $\square$

Remark 4.3. Corollary 4.2 is the precise form of the statement that RH is the marginal member of the family. The obstruction (7) — $\liminf \Psi = 0$, so that no approximation argument can establish $\Psi \geq 0$ — is the $\omega = 0$ shadow of a margin that is strictly positive and grows linearly in t for every $\omega > 0$. Whatever the difficulty of proving $\Psi_\omega \geq 0$ for a given $\omega \in (0, \frac{1}{2})$, it is not that the inequality is asymptotically saturated. Theorem 1.5 sharpens this into a dichotomy.

5. THE THRESHOLD

Proof of Theorem 1.4. Under $\vartheta \leq \omega$ we have $e^{-(\omega-\vartheta)t} \leq 1$ for $t \geq 0$, so Theorem 1.3 gives the uniform lower bound $\Psi_\omega(t) \geq \omega A(\omega)t + B(\omega) - A^*(\omega)$, and strict positivity for $t > t_0 = (A^* - B)/(\omega A)$, since $A(\omega) > 0$.

It remains to bound $A^* - B$. Write $z_1 = \gamma - i\omega$, $z_2 = \gamma + i\omega$. Then $z_1^2 + z_2^2 = 2(\gamma^2 - \omega^2)$ and $z_1^2 z_2^2 = (\gamma^2 + \omega^2)^2$, so the B -term of each zero is

$$\frac{\gamma^2 - \omega^2}{(\gamma^2 + \omega^2)^2} = \frac{z_1^2 + z_2^2}{2 z_1^2 z_2^2} = \frac{1}{2} \left[\frac{1}{z_1^2} + \frac{1}{z_2^2} \right],$$

while the A^* -term is $\frac{1}{2}(|z_1|^{-2} + |z_2|^{-2})$. For any $z = x + iy \neq 0$,

$$\frac{1}{|z|^2} - \operatorname{Re} \frac{1}{z^2} = \frac{(x^2 + y^2) - (x^2 - y^2)}{|z|^4} = \frac{2y^2}{|z|^4}.$$

Summing over the zeros (the imaginary parts cancel in conjugate pairs, so B equals the sum of the real parts) and noting $\operatorname{Im} z_1 = b - \omega$, $\operatorname{Im} z_2 = b + \omega$,

$$A^*(\omega) - B(\omega) = \sum_{\gamma} \left[\frac{(b - \omega)^2}{|\gamma - i\omega|^4} + \frac{(b + \omega)^2}{|\gamma + i\omega|^4} \right]. \quad (18)$$

Under $|b| \leq \vartheta \leq \omega$ we have $(b \mp \omega)^2 \leq 4\omega^2$ and $|\gamma \mp i\omega|^2 \geq a^2$, whence $A^* - B \leq 8\omega^2 \sum_{\gamma} a^{-4} = 8\omega^2 \sum_{\rho} (\operatorname{Im} \rho)^{-4}$. Computing the first 120 ordinates and estimating the tail with the Riemann–von Mangoldt formula (with the explicit error bound of [20], whose contribution is far below the last digit retained) gives $\sum_{\rho} (\operatorname{Im} \rho)^{-4} = 7.4346 \dots \cdot 10^{-5} < 7.44 \cdot 10^{-5}$. Together with $\omega A(\omega) = (\xi'/\xi)(\frac{1}{2} + \omega) \geq (2 + \gamma_0 - \log 4\pi) \omega$ from Corollary 4.2,

$$t_0(\omega) \leq \frac{8 \cdot 7.44 \cdot 10^{-5}}{0.0461914} \omega = 0.01288 \dots \times \omega,$$

which is less than 0.0065 throughout $0 < \omega < \frac{1}{2}$. $\square$

Remark 5.1. Two cheaper routes to non-strict positivity should be set beside Theorem 1.4, both already in [17]. Under RH, $\Psi \geq 0$ pointwise, and every term of the definition (8) is then non-negative, so $\Psi_{\omega} \geq 0$ everywhere — indeed $\Psi_{\omega}(t) > 0$ for $t > 0$, since $\Psi(t) > 0$ for $t \neq 0$ [17, Thm. 1.7]. More generally, under $\vartheta \leq \omega$ alone, Suzuki’s Herglotz argument (§1.4 above) gives $\Psi_{\omega} \geq 0$ everywhere via Kreĭn–Langer. What Theorem 1.4 adds is quantitative and elementary: an explicit linear lower bound with an explicit onset, obtained without the Kreĭn–Langer correspondence. In particular the criterion, wherever it holds, holds *with room to spare* — there is no analogue at $\omega > 0$ of the saturation (7).

Remark 5.2. The contrapositive is a single-point detector: a single verified value $\Psi_{\omega}(t^*) \leq 0$ at any $t^* > 0.0129\omega$ certifies a zero with $\operatorname{Re} \rho > \frac{1}{2} + \omega$ (by Suzuki’s pointwise statement, any $t^* > 0$ in fact suffices; the present form has the advantage of an explicit quantitative margin below which Ψ_{ω} cannot dip). Proposition 1.7 quantifies the rate at which such values appear when a violating zero exists. Proposition 1.8 shows, however, that no feasible computation can reach the regime where such values could occur.

Remark 5.3. The threshold $t_0(\omega) \leq 0.0129\omega$ tends to 0 with ω , and covers the boundary case $\vartheta = \omega$, where the oscillation is undamped: the reason a bounded, non-decaying oscillation cannot break the criterion is visible in (18) — a zero at height τ on the boundary $\operatorname{Re} \rho = \frac{1}{2} + \omega$ contributes oscillation of amplitude $\approx 2/\tau^2$, but contributes $\approx 4/\tau^2$ to the intercept $B(\omega)$; its own mean covers its own oscillation twice over. Only zeros *beyond* the half-plane can drive Ψ_{ω} negative for large t , which is Proposition 1.7.

6. THE TRANSITION AT ω_c

Write $\omega_c := \Theta - \frac{1}{2} = \vartheta$, so that Theorem 1.1 reads: the criterion at level ω holds if and only if $\omega \geq \omega_c$. Theorems 1.3 and 1.4 describe Ψ_{ω} at and above ω_c ; Theorem 1.5 adds the description below, by the same Laplace-transform mechanism that proves the sufficiency direction of Theorem 1.1.

Proof of Theorem 1.5. Part (i) is Theorem 1.3: for $\vartheta \leq \omega$ the remainder is bounded by $A^*(\omega)$.

For part (ii), suppose $\omega < \vartheta$ and, for a contradiction, that there are constants $C, D > 0$ and $0 \leq \varepsilon < \vartheta - \omega$ with $\Psi_{\omega}(t) \geq -Ce^{\varepsilon t} - D$ for all $t \geq 0$; set $f := \Psi_{\omega} + Ce^{\varepsilon t} + D \geq 0$. From (17),

$$|\Psi_{\omega}(t)| \leq \omega A(\omega) t + |B(\omega)| + A^*(\omega) e^{(\vartheta - \omega)t},$$

an estimate that requires no ordering of ϑ and ω (and A^* converges unconditionally by Lemma 2.1). Hence the Laplace transform $F(z) = \int_0^\infty f(t)e^{izt} dt$ converges, and is analytic, for $\text{Im } z > \vartheta - \omega$, and equals there

$$F(z) = -\frac{1}{z^2} \left(\frac{\xi'}{\xi} \right) \left(\frac{1}{2} + \omega - iz \right) + \frac{iC}{z - i\varepsilon} + \frac{iD}{z} : \quad (19)$$

on $\text{Im } z > \frac{1}{2} - \omega$ this is (9) together with two elementary transforms, and the identity extends to $\text{Im } z > \vartheta - \omega$ by analytic continuation. Let $\sigma_c \leq \vartheta - \omega$ denote the abscissa of convergence of F . Since $f \geq 0$, Landau's theorem [22, Ch. II, Thm. 5b] makes $i\sigma_c$ a singularity of F : no analytic continuation to a neighbourhood of $i\sigma_c$ exists. Now the right-hand side of (19) is meromorphic on $\mathbb{C}$, and its only possible singularities on the ray $\{iy : y > \varepsilon\}$ are poles of (ξ'/ξ) at real zeros $\frac{1}{2} + \omega + y \in (\frac{1}{2}, 1]$ — and there are none. Hence $\sigma_c \leq \varepsilon$, so F is analytic on $\text{Im } z > \varepsilon$ and coincides there with (19) by analytic continuation; consequently (19) has no poles with $\text{Im } z > \varepsilon$, i.e. every zero satisfies $\text{Re } \rho \leq \frac{1}{2} + \omega + \varepsilon$. Since $\omega + \varepsilon < \vartheta$, this contradicts the definition of ϑ . Therefore no such C, D exist, which is the $\liminf$ statement. The $\limsup$ statement follows from the same argument applied to $f := Ce^{\varepsilon t} + D - \Psi_\omega$, whose transform is (19) with the sign of the first term reversed. $\square$

Proof of Corollary 1.6. If $\Theta \leq \frac{1}{2} + \omega$, boundedness of $\Psi_\omega - (\xi'/\xi)(\frac{1}{2} + \omega)t$ is Theorem 1.5(i). Conversely, if it is bounded then $e^{-\varepsilon t}\Psi_\omega(t) \rightarrow 0$ for every $\varepsilon > 0$, so (ii) fails, forcing $\vartheta \leq \omega$. For the order parameter: on $[\omega_c, \frac{1}{2}]$, m equals the slope by (i); for $\omega < \omega_c$, part (ii) with any $0 < \varepsilon < \omega_c - \omega$ yields points $t_n \rightarrow \infty$ with $\Psi_\omega(t_n) \leq -e^{\varepsilon t_n}$, so $\Psi_\omega(t_n)/t_n \rightarrow -\infty$ and $m(\omega) = -\infty$. If $\omega_c > 0$, then $m(\omega_c) = (\xi'/\xi)(\frac{1}{2} + \omega_c) \geq (2 + \gamma_0 - \log 4\pi)\omega_c > 0$ by Corollary 4.2, while $m = -\infty$ immediately below: a jump. Under RH, $\omega_c = 0$ and $m(\omega) = (\xi'/\xi)(\frac{1}{2} + \omega) \rightarrow 0$ as $\omega \rightarrow 0^+$, matching $\liminf \Psi(t)/t = 0$ at $\omega = 0$ (Ψ is bounded under RH). $\square$

Remark 6.1. Below ω_c the oscillation is genuinely two-sided and of exponential rate arbitrarily close to $\omega_c - \omega$; at and above ω_c the function is linear plus $O(1)$. In the language of phase transitions, $m(\omega)$ is an order parameter and RH asserts a second-order (continuous) transition; any $\omega_c > 0$ would make it first-order, with “latent margin” $(\xi'/\xi)(\frac{1}{2} + \omega_c) \approx 0.0462\omega_c$. For ζ the proof of part (ii) uses only the absence of real zeros in $(\frac{1}{2}, 1]$; for Dirichlet L -functions the same argument is sensitive to real zeros, and Section 9 turns that sensitivity into a statement.

7. AN UNCONDITIONAL WINDOW

Proof of Proposition 1.8. Split (17) at the verification height: by [16], every zero with $|\text{Im } \rho| \leq T_v = 3 \cdot 10^{12}$ has $\text{Re } \rho = \frac{1}{2}$, i.e. the corresponding γ are real. For those zeros the summand of (17) is bounded by $(\gamma^2 + \omega^2)^{-1}$, and since the omitted high zeros have $\text{Re}(\gamma^2 + \omega^2) > 0$ (Lemma 2.1), the partial sum is at most $A(\omega)$; their total contribution to R_ω is at most $A(\omega)e^{-\omega t}$ in absolute value. For a zero with $|\text{Im } \rho| = |a| > T_v$, the explicit zero-free regions of [14] give $|b| \leq \frac{1}{2} - \nu(\log |a|)$ with ν as in the statement (both regions are valid, so their maximum is), and $|\gamma \pm i\omega|^2 \geq a^2$: its contribution is at most $e^{-\omega t} e^{(\frac{1}{2} - \nu(\log |a|))t} a^{-2}$. Summing against the Riemann–von Mangoldt density $dN(u) = (2\pi)^{-1} \log(u/2\pi) du$ and substituting $L = \log u$ gives $|R_\omega(t)| \leq A(\omega)e^{-\omega t} + e^{(\frac{1}{2} - \omega)t} E(t)$; the explicit error bound of [20] for the zero-counting function perturbs $E(t)$ by less than 10^{-23} , which is absorbed in every stated numerical value. At $t = 0$,

$$E(0) = S = 2 \int_{T_v}^\infty \frac{dN(u)}{u^2} = \frac{1}{\pi} \cdot \frac{\log(T_v/2\pi) + 1}{T_v} + O\left(\frac{\log T_v}{T_v^2}\right) < 3.2 \cdot 10^{-12},$$

and E is decreasing in t . The displayed lower bound for Ψ_ω follows from Theorem 1.3's main term; the window endpoints in Table 6 are the two roots of the right-hand side, the lower at $t_- \approx (1 - B/A)/(2\omega) < 0.0017\omega$.

For the endpoint law, note that the two branches of ν cross at $L_x = 8928.2$, and that for $L \leq L_x$ one has $\nu(L) = 1/(cL)$ with $c = 5.558691$. By the Laplace method, the exponent $-L - t/(cL)$ is maximised at $L^* = \sqrt{t/c}$ with value $-2\sqrt{t/c}$; splitting off $e^{-L/\sqrt{t}}$ to control the polynomial factor shows $E(t) = \exp(-2\sqrt{t/c}(1 + o(1)))$, valid while $L^* \leq L_x$, i.e. $t \leq cL_x^2 \approx 4.4 \cdot 10^8$ — far beyond every entry of Table 6. Solving $(\frac{1}{2} - \omega)t_+ = 2\sqrt{t_+/c}(1 + o(1))$ gives $t_+ = (4/c + o(1))(\frac{1}{2} - \omega)^{-2}$; numerically the law is accurate to 1% already at $\omega = 0.499$ ($0.7196 \cdot 10^6$ predicted against 726 493 computed). Beyond the crossover — relevant only once $\frac{1}{2} - \omega \lesssim 2 \cdot 10^{-5}$ — the Korobov–Vinogradov branch takes over and the same computation gives $t_+ \asymp (\frac{1}{2} - \omega)^{-5/2+o(1)}$. $\square$

Remark 7.1. Proposition 1.8 delimits what computation can say about the criterion. In the variable $x = e^t$ of the underlying prime sums, the window reaches $x \approx 10^{28}$ at $\omega = 0.1$, $x \approx 10^{48}$ at $\omega = \frac{1}{4}$, $x \approx 10^{131}$ at $\omega = 0.4$, and $x \approx 10^{3154}$ at $\omega = 0.49$: any computation of Ψ_ω at feasible arguments takes place deep inside a region where its positivity is already unconditionally known, and therefore carries no information about Θ beyond the inputs it is built on. In particular, “compute Ψ_ω at large t and watch for a dip” is not a probe of Θ . The two inputs act separably: the verification height sets the *level* of the obstruction bound, so the window grows with T_v like $t_+ \sim \frac{2 \log T_v}{1-2\omega}$ at moderate ω , while the zero-free region sets its *decay*, which is invisible at moderate ω (a 1–7% enlargement of t_+ for $\omega \leq 0.45$) and dominant at the PNT end, where it produces the endpoint law $t_+ = (4/c + o(1))(\frac{1}{2} - \omega)^{-2}$ and reconnects the family continuously to the unconditional case $\omega \geq \frac{1}{2}$. This is the honest quantitative content available to the criterion short of a genuine zero-free half-plane.

8. THE RATE AT WHICH A VIOLATING ZERO IS DETECTED

Proof of Proposition 1.7. By (11) the quartet $\{\rho_0, \overline{\rho_0}, 1 - \rho_0, 1 - \overline{\rho_0}\}$ corresponds to $\gamma \in \{\pm\tau_0 \pm i\delta\}$. Fix such a $\gamma = a + i\delta$ with $a = \pm\tau_0$. In (17), the term $e^{-i\gamma t} = e^{-iat}e^{\delta t}$ dominates its partner $e^{i\gamma t} = e^{iat}e^{-\delta t}$ as $t \rightarrow \infty$, so the contribution of this γ to the oscillatory part is

$$-\frac{e^{(\delta-\omega)t}e^{-iat}}{2(\gamma - i\omega)^2}(1 + O(e^{-2\delta t})), \quad \gamma - i\omega = a + i(\delta - \omega).$$

Summing over $a = \pm\tau_0$ and adding the contribution from the two γ with $\text{Im } \gamma = -\delta$ (for which the dominant term is $e^{i\gamma t}$, with denominator $(\gamma + i\omega)^2$) yields

$$-e^{(\delta-\omega)t} \operatorname{Re} \left[\frac{e^{-i\tau_0 t}}{(\tau_0 + i(\delta - \omega))^2} + \frac{e^{i\tau_0 t}}{(\tau_0 - i(\delta - \omega))^2} \right] (1 + o(1)) = -\frac{2e^{(\delta-\omega)t}}{\tau_0^2 + (\delta - \omega)^2} \cos(\tau_0 t + \alpha)(1 + o(1)),$$

where $\alpha = 2 \arg(\tau_0 + i(\delta - \omega))$. The non-oscillatory part of the quartet's contribution is $\omega A_0 t + B_0 = O(t)$ with A_0, B_0 the corresponding partial sums in (10). The lim sup statement follows since $\cos(\tau_0 t + \alpha)$ attains ± 1 on a relatively dense set. $\square$

Remark 8.1. Proposition 1.7 should be contrasted with the Nyman–Beurling criterion, in which RH is equivalent to $d_N \rightarrow 0$ for an explicit sequence of Hilbert-space distances with $\liminf d_N^2 \log N \geq \sum_\rho m_\rho^2 |\rho|^{-2}$ [1, 6]. There, by an identity of Balazard–Saias–Yor [2], the natural size of the violation signal of a zero $\rho = \frac{1}{2} + \delta + i\tau$ is $\log |\rho/(1-\rho)| \approx \delta/(\frac{1}{4} + \tau^2)$, to be seen against a baseline of order $1/\log N$: detection requires $\log N \gtrsim \tau^2/\delta$, i.e. $N \geq \exp(c\tau^2/\delta)$.

Here, in the same variable $t = \log x$, the same zero drives Ψ_ω with an envelope growing like $e^{(\delta-\omega)t}$ and explicit amplitude: combining Proposition 1.7 with Corollary 4.2, the height at which a violating quartet at fixed τ_0 overtakes the margin is $t \asymp (\delta - \omega)^{-1} \log((\delta - \omega)^{-1})$. The criterion is exponentially sensitive where Nyman–Beurling is nearly blind — though Proposition 1.8 shows this sensitivity, too, begins only beyond every feasible argument, since a violating zero must lie above the verification height.

9. DIRICHLET L -FUNCTIONS, AND THE EXTENDED SELBERG CLASS

Everything so far used four inputs: the Hadamard expansion, the functional-equation antisymmetry of ξ'/ξ about $s = \frac{1}{2}$, the strip bound $|\operatorname{Im} \gamma| \leq \frac{1}{2}$, and the absence of real zeros in $(\frac{1}{2}, 1]$. All four are available for real primitive Dirichlet characters, so the theory transfers whole. What changes is instructive: the margin grows with the conductor, and the absence-of-real-zeros input — free for ζ — becomes both necessary and visible.

Let χ be a real primitive character mod q , $a = (1 - \chi(-1))/2$, and

$$\xi(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma\left(\frac{s+a}{2}\right) L(s, \chi),$$

entire of order one with $\xi(s, \chi) = \xi(1-s, \chi)$, since the root number of a real primitive character equals $+1$ by Gauss’s theorem (see [7, §9]). Because the coefficients are real, zeros also pair under conjugation, so the zero set of $\Xi_\chi(z) = \xi(\frac{1}{2} - iz, \chi)$ carries the full symmetry $\gamma \mapsto -\gamma$, $\gamma \mapsto \bar{\gamma}$ used throughout, and $\Psi_{\omega, \chi}$ is defined by the series (13) over these γ . We restrict to $q \leq 4 \cdot 10^5$, where the numerical verification of GRH by Platt [15] guarantees that $L(s, \chi)$ has no real zeros in $(0, 1)$ and no non-real zeros of height below $\max(10^8/q, 200)$; this replaces the two numerical inputs used for ζ .

Theorem 9.1. *Let χ be real primitive of modulus $q \leq 4 \cdot 10^5$ and $0 < \omega < \frac{1}{2}$, with sums over the zeros γ of Ξ_χ .*

- (i) *Unconditionally, $A_\chi(\omega) := \sum_\gamma (\gamma^2 + \omega^2)^{-1} = \omega^{-1}(\xi'/\xi)(\frac{1}{2} + \omega, \chi)$ and $B_\chi(\omega) := \sum_\gamma (\gamma^2 - \omega^2)(\gamma^2 + \omega^2)^{-2} = (\xi'/\xi)'(\frac{1}{2} + \omega, \chi)$; moreover $(\xi'/\xi)(\frac{1}{2}, \chi) = 0$, so the margin slope vanishes at $\omega = 0$ exactly as for ζ .*
- (ii) *If $\vartheta_\chi := \sup_\rho \operatorname{Re}(\rho) - \frac{1}{2} \leq \omega$ then*

$$\Psi_{\omega, \chi}(t) = \left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega, \chi\right) t + \left(\frac{\xi'}{\xi}\right)' \left(\frac{1}{2} + \omega, \chi\right) + O\left(A_\chi^*(\omega) e^{-(\omega - \vartheta_\chi)t}\right),$$

and $\Psi_{\omega, \chi}(t) > 0$ for every $t > (A_\chi^(\omega) - B_\chi(\omega))/(\omega A_\chi(\omega))$, with $A_\chi^* - B_\chi \leq 8\omega^2 \sum_\rho (\operatorname{Im} \rho)^{-4}$.*

- (iii) *The analogue of Theorem 1.5 holds verbatim.*

Proof. The proofs of Sections 2–6 apply with $\xi(\cdot)$ replaced by $\xi(\cdot, \chi)$. Oddness of $(\xi'/\xi)(\cdot, \chi)$ about $\frac{1}{2}$ follows from the even functional equation, which gives $(\xi'/\xi)(\frac{1}{2}, \chi) = 0$ and removes the constant from the Hadamard expansion of Ξ'_χ/Ξ_χ , exactly as in [17, §2.2]. The strip bound $|\operatorname{Im} \gamma| \leq \frac{1}{2}$ holds as for ζ . By [15] there are no real zeros in $(0, 1)$ (so no purely imaginary γ , and no poles of $(\xi'/\xi)(\cdot, \chi)$ on $(\frac{1}{2}, 1]$, which part (iii) needs), and every non-real zero off the critical line has height above 200, so $\operatorname{Re}(\gamma^2 + \omega^2) > 0$ for every γ . The pole terms $1/s + 1/(s-1)$ are absent throughout, since $\xi(s, \chi)$ is entire. $\square$

Table 1 records the margin across conductors: $B_\chi(0) = (\xi'/\xi)'(\frac{1}{2}, \chi)$ — the analogue of Remark 3.1 — and the slope at $\omega = \frac{1}{4}$. The margin grows with the conductor, because the lowest zero descends: *within this family, ζ is the most marginal member.*

The arithmetic side mirrors (14), with the pole block absent.

L -function	$B(0)$	slope $(\xi'/\xi)(\frac{3}{4}, \cdot)$
ζ	0.046210	0.011552
$\chi \bmod 3$	0.113402	0.028340
$\chi \bmod 4$	0.156030	0.038978
$\chi \bmod 5$	0.156914	0.039206

TABLE 1. The margin across conductors.

Proposition 9.2. *Let χ, q, a be as above and $\alpha := (\frac{1}{2} + \omega + a)/2$. For $t > 0$,*

$$\Psi_{\omega, \chi}(t) = \frac{t}{2} \left[\psi(\alpha) + \log \frac{q}{\pi} \right] + \frac{1}{4} \left[\psi^{(1)}(\alpha) - e^{-2\alpha t} \Phi(e^{-2t}, 2, \alpha) \right] - \sum_{n \leq e^t} \frac{\Lambda(n) \chi(n)}{n^{\frac{1}{2} + \omega}} (t - \log n).$$

Proof. By uniqueness of Fourier transforms it suffices to show that the right-hand side has Laplace transform $-z^{-2}(\xi'/\xi)(\frac{1}{2} + \omega - iz, \chi)$ for $\text{Im } z$ large, since the series defining $\Psi_{\omega, \chi}$ has this transform by the argument of [17, §§2.2, 11], using $(\xi'/\xi)(\frac{1}{2}, \chi) = 0$. Write $s = \frac{1}{2} + \omega - iz$; for $\text{Re } s > 1$,

$$\left(\frac{\xi'}{\xi} \right)(s, \chi) = \frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi\left(\frac{s+a}{2}\right) - \sum_{n \geq 2} \Lambda(n) \chi(n) n^{-s},$$

with no pole terms. The transforms $\int_0^\infty t e^{izt} dt = -z^{-2}$ and $\int_0^\infty (t - \log n) n^{-\frac{1}{2} - \omega} \mathbf{1}_{[\log n, \infty)}(t) e^{izt} dt = -z^{-2} n^{-s}$ handle the first and third pieces. For the middle piece we claim, for $\alpha > 0$ and $\text{Im } z > 0$,

$$\int_0^\infty \frac{1}{4} \left[\psi^{(1)}(\alpha) - e^{-2\alpha t} \Phi(e^{-2t}, 2, \alpha) \right] e^{izt} dt = -\frac{1}{2z^2} \left[\psi\left(\alpha - \frac{iz}{2}\right) - \psi(\alpha) \right]. \quad (20)$$

Indeed $\psi^{(1)}(\alpha) - e^{-2\alpha t} \Phi(e^{-2t}, 2, \alpha) = \sum_{k \geq 0} (k + \alpha)^{-2} (1 - e^{-2(k+\alpha)t})$, and per k , with $c = 2(k + \alpha)$, $\int_0^\infty (1 - e^{-ct}) e^{izt} dt = \frac{i}{z} - \frac{1}{c - iz}$; the identity

$$\frac{1}{4(k + \alpha)^2} \left[\frac{i}{z} - \frac{1}{2(k + \alpha) - iz} \right] = -\frac{1}{2z^2} \left[\frac{1}{k + \alpha} - \frac{1}{k + \alpha - iz/2} \right]$$

is elementary — both sides equal $\frac{i}{4z} [(k + \alpha)(k + \alpha - \frac{iz}{2})]^{-1}$ — and summing over k gives (20). Since $(s + a)/2 = \alpha - iz/2$, assembling the three pieces yields $-z^{-2}(\xi'/\xi)(s, \chi)$ for $\text{Im } z > \frac{1}{2}$, hence for $\text{Im } z > \frac{1}{2} - \omega$ by analytic continuation. Finally $\Psi_{\omega, \chi}(0) = 0$, since $\Phi(1, 2, \alpha) = \zeta(2, \alpha) = \psi^{(1)}(\alpha)$. $\square$

Table 2 tests Proposition 9.2 against the main term of Theorem 9.1(ii), using prime powers $p^m \leq e^{15}$ and no information about the zeros; the difference sits inside the envelope $A_\chi(\omega) e^{-\omega t}$ in every row and decays at the predicted rate. (The differences are roughly ten times the ζ case, as they must be: A_χ is about three times larger and the lowest zeros are lower — 6.02 for $q = 4$ against 14.13 — so the leading damped oscillation is bigger.)

The genuinely new phenomenon is what the criterion *cannot* see. A real zero enters the series with a locked phase, and pushes $\Psi_{\omega, \chi}$ up.

Proposition 9.3 (Blindness to real zeros). *Let χ be a real primitive character, $0 < \omega < \frac{1}{2}$, and assume $\frac{1}{2} + \omega$ is not a zero of $L(\cdot, \chi)$. Write $\beta_0 \in [\frac{1}{2}, 1)$ for the largest real zero of $L(\cdot, \chi)$ in $[\frac{1}{2}, 1)$, with $\beta_0 = \frac{1}{2}$ if there is none, and $\vartheta_{\text{nr}} := \sup\{\text{Re } \rho - \frac{1}{2} : \rho \text{ non-real}\}$.*

q	ω	t	$\Psi_{\omega,\chi}(t)$ (primes)	main term	difference	$A_\chi(\omega)e^{-\omega t}$
3	0.2	6	0.25378628	0.24936876	4.4×10^{-3}	3.4×10^{-2}
3	0.2	15	0.45339439	0.45344292	-4.9×10^{-5}	5.6×10^{-3}
3	0.3	15	0.62323052	0.62324617	-1.6×10^{-5}	1.3×10^{-3}
3	0.4	15	0.79281810	0.79282263	-4.5×10^{-6}	2.8×10^{-4}
4	0.2	6	0.35287485	0.34295358	9.9×10^{-3}	4.7×10^{-2}
4	0.2	15	0.62564343	0.62367332	2.0×10^{-3}	7.8×10^{-3}
4	0.3	15	0.85733527	0.85691254	4.2×10^{-4}	1.7×10^{-3}
4	0.4	15	1.08963532	1.08954490	9.0×10^{-5}	3.9×10^{-4}
5	0.2	6	0.36369455	0.34497036	1.9×10^{-2}	4.7×10^{-2}
5	0.2	15	0.62696177	0.62731266	-3.5×10^{-4}	7.8×10^{-3}
5	0.3	15	0.86199461	0.86206119	-6.7×10^{-5}	1.7×10^{-3}
5	0.4	15	1.09632744	1.09633964	-1.2×10^{-5}	3.9×10^{-4}

TABLE 2. Proposition 9.2 tested against the arithmetic side. No zeros are used.

- (i) A real zero pair $\{\frac{1}{2} + \delta, \frac{1}{2} - \delta\}$ with $\delta > \omega$ contributes

$$\frac{e^{(\delta-\omega)t}}{(\delta-\omega)^2} + \frac{e^{-(\delta+\omega)t}}{(\delta+\omega)^2} > 0$$

to the oscillatory part (17) of $\Psi_{\omega,\chi}$.

- (ii) If $\Psi_{\omega,\chi}(t) \geq 0$ for all sufficiently large t , then $\vartheta_{\text{nr}} \leq \max(\omega, \beta_0 - \frac{1}{2})$.
(iii) Conversely, suppose $\vartheta_{\text{nr}} \leq \max(\omega, \beta_0 - \frac{1}{2})$, that every non-real zero satisfies $|\text{Im } \rho| \geq \frac{1}{2}$, and, in the case $\beta_0 > \frac{1}{2} + \omega$, that $2 \sum_{\rho \text{ non-real}} (\text{Im } \rho)^{-2} < (\beta_0 - \frac{1}{2} - \omega)^{-2}$. Then $\Psi_{\omega,\chi}(t) \geq 0$ for all sufficiently large t .

For $q \leq 4 \cdot 10^5$ the side conditions in (iii) hold by [15] (indeed there are no real zeros at all), so (ii) and (iii) are an equivalence. In particular the shifted criterion cannot detect Siegel zeros, and the real-zero hypothesis in [18, Thm. 1.1] cannot be dropped.

Proof. (i) For $\gamma = \pm i\delta$ one has $e^{\mp i\gamma t} = e^{\pm \delta t}$ and $(\gamma \mp i\omega)^2 = -(\delta \mp \omega)^2$, so each of the four terms that the pair contributes to (17) is positive; summing them gives the display.

(ii) This is the Landau argument of Theorem 1.5(ii). Suppose $\Psi_{\omega,\chi} \geq 0$ eventually; its Laplace transform F converges in a half-plane and equals $-z^{-2}(\xi'/\xi)(\frac{1}{2} + \omega - iz, \chi)$ plus an entire correction. By Landau's theorem, $i\sigma_c$ is a singularity of F ; the only candidates on the closed positive imaginary axis are $z = 0$ and points $i(\beta - \frac{1}{2} - \omega)$ with β a real zero, so $\sigma_c \leq \max(0, \beta_0 - \frac{1}{2} - \omega)$. Since F is analytic above σ_c and agrees with the meromorphic expression, every zero satisfies $\text{Re } \rho - \frac{1}{2} - \omega \leq \sigma_c$, which is the claim.

(iii) Write $\Psi_{\omega,\chi} = \omega A_\chi t + B_\chi + R$ as in (17). If $\beta_0 \leq \frac{1}{2} + \omega$, all zeros satisfy $\text{Re } \rho \leq \frac{1}{2} + \omega$: real-zero pairs with $\delta \leq \omega$ contribute positively to A_χ (their terms are $(\omega^2 - \delta^2)^{-1} > 0$), non-real zeros contribute $\text{Re}[(\gamma^2 + \omega^2)^{-1}] > 0$ by $|\text{Im } \rho| \geq \frac{1}{2} \geq |b|$, so $A_\chi(\omega) > 0$, $|R| \leq A_\chi^* < \infty$, and the margin wins for large t . If $\beta_0 = \frac{1}{2} + \delta_r > \frac{1}{2} + \omega$, then by (i) the largest real pair contributes at least $e^{(\delta_r - \omega)t}(\delta_r - \omega)^{-2}(1 + o(1))$, all other real pairs contribute positively, and the non-real zeros contribute at most $e^{(\delta_r - \omega)t} \cdot 2 \sum_{\text{nr}} (\text{Im } \rho)^{-2}$ in absolute value, which the amplitude condition makes strictly smaller; the linear terms are $O(t)$. Hence $\Psi_{\omega,\chi} \rightarrow +\infty$. $\square$

Remark 9.4 (The extended Selberg class). Suzuki has extended the unshifted theory to the extended Selberg class $S^\sharp$ [18]: the screw function there acquires the extra terms $-iB_F t - \frac{m_0}{2} t^2$ (with $iB_F = (\xi'_F/\xi_F)(\frac{1}{2})$ and m_0 the order of the central zero), the criterion concerns $\operatorname{Re}(-g_F)$, is stated under the hypothesis that F has no real zeros off $s = \frac{1}{2}$, and takes the “eventually non-negative” form even without a shift — which makes an explicit threshold in the style of Theorem 1.4 more, not less, relevant there. Three remarks on how the present results sit in that generality. First, the functional equation $\xi_F(s) = \omega_F \xi_F(1 - \bar{s})$ forces $\operatorname{Re}(\xi'_F/\xi_F)(\frac{1}{2}) = 0$ for *every* $F \in S^\sharp$, so the margin slope $\operatorname{Re}(\xi'_F/\xi_F)(\frac{1}{2} + \omega)$ vanishes at $\omega = 0$ always: the marginality mechanism of Corollary 4.2 is universal, the non-self-dual case being handled through the pair $(F, \bar{F})$, which restores the $\pm\gamma$ symmetry used in the proofs. Second, a central zero of order m_0 contributes the non-negative term $m_0[t/\omega - (1 - e^{-\omega t})/\omega^2]$ to the shifted function, so families with forced central zeros are non-marginal at every $\omega > 0$, with slope blowing up like m_0/ω as $\omega \rightarrow 0^+$. Third, Proposition 9.3 is the quantitative reason for the real-zero hypothesis in [18, Thm. 1.1]: real zeros push the screw function up, at the fastest exponential rate present, and no criterion of this family can see them.

10. NUMERICAL VERIFICATION

All computations were done in `mpmath` at 20–30 decimal digits, with $\Lambda(n)$ obtained by sieving. Scripts are available from the author. The Dirichlet computations — functional-equation and reality checks for $\xi(\cdot, \chi)$, zero counts against the Riemann–von Mangoldt formula, the closed forms of Theorem 9.1(i) against direct zero sums, and the arithmetic side — are reported in Section 9 (Tables 1 and 2).

10.1. The arithmetic side. As a check on the implementation of (14) we reproduce a value printed by Suzuki. In the proof of [17, Thm. 4.1] he records $\Psi(t_2) = 0.0396618\dots$ at $t_2 = 0.464002\dots$. Evaluating (14) at $\omega = 0$ gives

$$\Psi(0.464002) = 0.03966175656\dots, \quad \Psi(0) = 0.$$

Separately, the per-zero identity between the definition (8) and the series (13), and the decomposition (16), were verified symbolically and to 30 digits at complex γ .

10.2. The closed forms of Proposition 1.2. Table 3 compares the closed forms against direct summation over the first 250 zeros together with a Riemann–von Mangoldt tail estimate $dN = (2\pi)^{-1} \log(\tau/2\pi) d\tau$. The relative discrepancy is 1.13×10^{-4} in every row, independent of ω ; this is the error of the tail estimate, not of the identity.

ω	$A(\omega)$ (sum + tail)	$A(\omega)$ (closed form)	$B(\omega)$ (sum + tail)	$B(\omega)$ (closed form)
0.10	0.04621447	0.04620924	0.04621299	0.04620776
0.25	0.04621057	0.04620534	0.04620128	0.04619605
0.40	0.04620333	0.04619810	0.04617957	0.04617434

TABLE 3. Proposition 1.2 against direct summation over 250 zeros plus tail.

10.3. The asymptotic, tested against the primes. Table 4 is the substantive check: the left column is computed from (14), i.e. from prime powers $p^m \leq e^{15}$ and elementary functions, using no information about the zeros whatsoever; the right column is the main term of Theorem 1.3. The difference lies inside the envelope $A(\omega)e^{-\omega t}$ of Theorem 1.3 in every row and decays at the predicted rate.

ω	t	$\Psi_\omega(t)$ from (14)	main term	difference	$A(\omega)e^{-\omega t}$
0.20	6	0.10222000	0.10164948	5.7×10^{-4}	1.4×10^{-2}
0.20	15	0.18501313	0.18482211	1.9×10^{-4}	2.3×10^{-3}
0.30	6	0.12966687	0.12935586	3.1×10^{-4}	7.6×10^{-3}
0.30	15	0.25414919	0.25410476	4.4×10^{-5}	5.1×10^{-4}
0.40	6	0.15721911	0.15704977	1.7×10^{-4}	4.2×10^{-3}
0.40	15	0.32337324	0.32336293	1.0×10^{-5}	1.2×10^{-4}

TABLE 4. Theorem 1.3 tested against the arithmetic side. No zeros are used.

10.4. The margin and the threshold. Table 5 records the slope, the intercept, the ratio $r := B/A$, the threshold $t_0 = (A - B)/(\omega A) = (1 - r)/\omega$ that Theorem 1.4 yields in the RH case ($A^* = A$), and the bound $8\omega^2 \sum_\rho (\text{Im } \rho)^{-4}/(\xi'/\xi)(\frac{1}{2} + \omega)$ valid for every $\vartheta \leq \omega$. The row $\omega = 0$ returns 1.2×10^{-30} for the slope, i.e. zero to working precision, confirming $(\xi'/\xi)(\frac{1}{2}) = 0$.

ω	$(\xi'/\xi)(\frac{1}{2} + \omega)$	$B(\omega)$	$r(\omega)$	t_0 (RH case)	t_0 bound (any $\vartheta \leq \omega$)
0.49	0.0226342	0.0461565	0.99922857	0.00157	0.00631
0.40	0.0184792	0.0461743	0.99948567	0.00129	0.00515
0.30	0.0138610	0.0461899	0.99971057	0.00096	0.00386
0.20	0.0092414	0.0462011	0.99987132	0.00064	0.00257
0.10	0.0046209	0.0462078	0.99996782	0.00032	0.00129
0.05	0.0023105	0.0462094	0.99999196	0.00016	0.00064
0.01	0.00046210	0.0462100	0.99999968	3.2×10^{-5}	1.3×10^{-4}
0.001	0.000046210	0.0462100	0.9999999968	3.2×10^{-6}	1.3×10^{-5}
0	1.2×10^{-30}	0.0462100	—	—	—

TABLE 5. The margin of Theorem 1.3 and the thresholds of Theorem 1.4. $1 - r(\omega) \approx 0.0032\omega^2$, reflecting $A - B = 2\omega^2 \sum_\gamma (\gamma^2 + \omega^2)^{-2}$.

10.5. The unconditional window. Table 6 records the window of Proposition 1.8: the two roots of $\omega At + B - Ae^{-\omega t} - e^{(\frac{1}{2} - \omega)t} E(t)$, with, for comparison, the endpoint that the undamped bound $E \equiv S = 2.96 \cdot 10^{-12}$ would give. The zero-free damping is a 1–7% effect for $\omega \leq 0.45$ and dominant near $\omega = \frac{1}{2}$, where the endpoint law $t_+ \approx 0.7196 (\frac{1}{2} - \omega)^{-2}$ takes over (predicted $7.20 \cdot 10^5$ against computed 726 493 at $\omega = 0.499$).

ω	$t_-(\omega)$	$t_+ (E \equiv S)$	$t_+(\omega)$	$x_+ = e^{t_+}$
0.05	8.0×10^{-5}	55.06	55.84	$10^{24.3}$
0.10	1.6×10^{-4}	63.67	64.68	$10^{28.1}$
0.20	3.2×10^{-4}	87.98	89.86	$10^{39.0}$
0.25	4.0×10^{-4}	107.19	109.94	$10^{47.8}$
0.30	4.8×10^{-4}	136.02	140.42	$10^{61.0}$
0.40	6.4×10^{-4}	282.06	300.89	$10^{130.7}$
0.45	7.2×10^{-4}	580.82	663.59	$10^{288.2}$
0.49	7.9×10^{-4}	3079.1	7262.3	10^{3154}
0.494	7.9×10^{-4}	—	20 178	10^{8763}
0.497	8.0×10^{-4}	—	80 694	$10^{35\,050}$
0.499	8.0×10^{-4}	—	726 493	$10^{315\,500}$

TABLE 6. Proposition 1.8: unconditional positivity of Ψ_ω on (t_-, t_+) , from RH verified to $3 \cdot 10^{12}$ [16] and the zero-free regions of [14].

10.6. Sensitivity. Table 7 measures the growth rate of the envelope of a synthetic violating quartet at ordinate $\tau_0 = 50$, taking the maximum of $|\Psi_\omega^{(\rho_0)}|$ over one period of $\cos(\tau_0 t)$. Where the exponential has fully separated from the linear drift the measured rate agrees with $\delta - \omega$ to five digits; where it has not, the drift contributes and the measured rate is smaller. Rows with $\delta < \omega$ grow at the rate of the linear drift $\omega A_0 t + B_0$ alone: the measured rates 0.01833 and 0.01638 match $\frac{1}{20} \log((60\omega A_0 + B_0)/(40\omega A_0 + B_0)) = 0.01838$ and 0.01682 to within the residual damped oscillation, i.e. no exponential component survives, as predicted. Comparison against the amplitude $2(\tau_0^2 + (\delta - \omega)^2)^{-1} e^{(\delta - \omega)t}$ of Proposition 1.7 gives agreement to 1–6% at the heights tested, the discrepancy being the $O(t)$ term.

δ	ω	envelope at $t = 40$	envelope at $t = 60$	measured rate	$\delta - \omega$
0.30	0.10	2.417	1.3195×10^2	+0.20000	+0.2000
0.20	0.10	5.175×10^{-2}	3.341×10^{-1}	+0.09325	+0.1000
0.15	0.10	1.392×10^{-2}	2.727×10^{-2}	+0.03364	+0.0500
0.10	0.20	1.445×10^{-2}	2.084×10^{-2}	+0.01833	−0.1000
0.05	0.10	8.114×10^{-3}	1.126×10^{-2}	+0.01638	−0.0500

TABLE 7. Proposition 1.7: envelope growth for a synthetic quartet at $\tau_0 = 50$.

11. WHAT THIS SUPPLIES, AND WHAT IT DOES NOT

11.1. The shape of the problem. Theorem 1.3 exhibits the criterion as a competition:

$$\Psi_\omega(t) = \underbrace{\left(\frac{\xi'}{\xi}\right)\left(\frac{1}{2} + \omega\right)t + \left(\frac{\xi'}{\xi}\right)' \left(\frac{1}{2} + \omega\right)}_{\text{margin, linear in } t, \text{ unconditional}} - \underbrace{e^{-\omega t} \times (\text{oscillation over zeros})}_{\text{obstruction}}. \quad (21)$$

Under Hypothesis $\mathbf{H}(\vartheta)$ with $\vartheta \leq \omega$ the obstruction is bounded by $A^*(\omega)e^{-(\omega - \vartheta)t}$ and the margin wins beyond an explicit $t_0(\omega) \leq 0.0129\omega$. If instead there is a zero at $\text{Re } \rho = \frac{1}{2} + \delta$ with $\delta > \omega$, the obstruction contains a term of size $e^{(\delta - \omega)t}$ and eventually swamps the margin.

Theorem 1.1 is exactly this dichotomy; Theorems 1.3–1.4 and Proposition 1.7 make both sides of it quantitative.

11.2. No free lunch. To prove $\Psi_\omega \geq 0$ for a fixed $\omega < \frac{1}{2}$ from the arithmetic side one must show that the fluctuation of

$$\sum_{n \leq e^t} \frac{\Lambda(n)}{n^{\frac{1}{2} + \omega}} (t - \log n)$$

about its smooth part is bounded by the linear margin $(\xi'/\xi)(\frac{1}{2} + \omega)t \approx 0.0462\omega t$. Without zero-free information that fluctuation may a priori be as large as $e^{(\frac{1}{2} - \omega)t}$. Converting an exponential a priori bound into a linear one is precisely the content of a zero-free half-plane. The reformulation transports the difficulty faithfully; it does not reduce it, and we make no claim that it does.

11.3. Three respects in which the shifted case is nevertheless better behaved. (i) *The criterion is not saturated.* For $\omega = 0$ the inequality $\Psi \geq 0$ has $\liminf \Psi = 0$ conditionally (7), so no argument that establishes the inequality up to a positive error can succeed. Corollary 4.2 shows that this obstruction is a property of the point $\omega = 0$ alone: it is the vanishing of ξ'/ξ at the centre of the critical strip. For every $\omega > 0$ there is a margin, it is explicit, and it grows without bound.

(ii) *The required cancellation is linear, not square-root.* Most positivity criteria for RH demand square-root cancellation in an arithmetic sum. Here one needs only $o(t)$ control of a twice-integrated weighted Chebyshev error — a much weaker requirement, bought by asking for a weaker conclusion.

(iii) *Failure is exponentially visible.* Proposition 1.7 and Remark 8.1.

11.4. Zero-density estimates: what they can and cannot do here. The obstruction in (21) is, by (17) and (11), a sum over zeros weighted by $e^{(\operatorname{Re} \rho - \frac{1}{2} - \omega)t}$ and $|\rho|^{-2}$. Sums of this shape are what zero-density theorems — from Ingham to the recent improvement of Guth and Maynard [10] — are built to control, so it is natural to ask whether, for some $\omega < \frac{1}{2}$, known density estimates inserted into (17) bound the obstruction by the margin. The answer is no, for every ω , and for a structural reason worth recording: density estimates bound *counts* $N(\sigma, T)$, and every known bound is consistent with the existence of a single zero at any fixed $\beta < 1$; a single permitted zero at $\beta > \frac{1}{2} + \omega$ contributes $e^{(\beta - \frac{1}{2} - \omega)t}$ and defeats any linear margin as $t \rightarrow \infty$. No argument using count upper bounds alone can close the gap — to bound the obstruction by the margin for all t is to assert the zero-free half-plane.

What density and zero-free-region technology does control is the *onset*: it constrains where in t a violation could first appear. The Korobov–Vinogradov region forces a zero at β close to 1 to have height $\tau \geq \exp(c(1 - \beta)^{-3/2})$, and a zero of height τ enters (17) with weight τ^{-2} ; both effects postpone the possible failure of $\Psi_\omega \geq 0$ to explicit ranges of t , exactly as verification to height T_v does in Proposition 1.8. In this precise sense the criterion’s open content lives entirely in the tail $t \rightarrow \infty$, where only genuine zero-free-half-plane information can reach. The productive question is therefore not whether density estimates prove the criterion — they cannot — but how the two available inputs shape the unconditional window of Proposition 1.8, and there the answer is now quantitative. The verification height sets the *level* of the obstruction bound: $t_+ \approx (\frac{1}{2} - \omega)^{-1} \log(1/S)$ at moderate ω , so raising T_v moves the window logarithmically. The zero-free region sets its *decay*: negligible at moderate ω , it takes over at the PNT end and produces the endpoint law $t_+ = (4/c + o(1))(\frac{1}{2} - \omega)^{-2}$, so the

constant c of [14] enters the criterion quadratically. All the action of zero-free technology on this family is concentrated near $\omega = \frac{1}{2}$.

11.5. Status of the family. We note finally that Ψ_ω appears to be unworked. As of August 2026, Suzuki’s paper has twelve recorded citations (Semantic Scholar), of which eight are his own; none of them takes up Ψ_ω or Theorem 1.1. In particular the later work extending screw functions to the extended Selberg class [18] contains no shifted family — the symbol ω there is the root number — and its “eventually non-negative” criterion, together with its real-zero hypothesis, is exactly the point at which the shifted, quantitative theory of the present paper attaches (Section 9).

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